

→ How camera sees 3D

> A camera converts a point in 3D space into a point on a 2D image

$P(X, Y, Z)$

↳ distance in front of the camera (increasing Z moves image closer to the center)

⇒ in an ideal pinhole camera:

$$x = \frac{X}{Z} \quad y = \frac{Y}{Z}$$

$(1,1,2) ; (2,2,4) ; (3,3,6)$) projects all to the same image position

1. If an object keeps the same X and Y , but its Z doubles, what happens to its displacement from the image center?
2. Why can't one ordinary image tell us the exact distance to an arbitrary object?
3. If a point is directly in front of the camera, what would you expect about its X and Y coordinates?

1. comes closer 2x to the center

2. because the image is 2D but the taken world is 3D

3. 0, 0

> pixel (0,0) is image's top-left corner

the optical center is: (c_x, c_y) → principal point - where the camera's optical axis intersects the image plane

⇒ to convert the normalized projection into pixels:

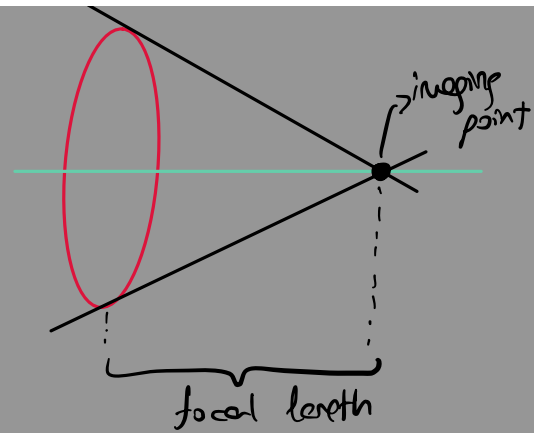
$$u = f_x \frac{X}{Z} + c_x$$

focal length

$$v = f_y \frac{Y}{Z} + c_y$$

Here:

- u, v : pixel coordinates
- f_x, f_y : focal lengths measured in pixels
- c_x, c_y : principal point
- X, Y, Z : point coordinates relative to the camera



1. If $X = 0$ and $Y = 0$, what pixel coordinates (u, v) do the equations produce?
2. If focal length increases while everything else stays fixed, does the object appear closer to or farther from the image center?
3. Suppose $f_x = 800$, $c_x = 640$, $X = 0.5$, and $Z = 2$. What is u ?
4. Without recalculating from scratch, what should happen to u if Z doubles while X stays fixed?

1. $u = c_x$; $v = c_y$

2. further.

3. $800 \cdot \frac{0.5}{2} + 640 = 840$

4. decrease

focal length

> cam might have focal length in millimeters

> but img coords. are pixels

> so camera geometry uses f_x, f_y measured in pixels.

$$f_x = \frac{f}{s_x}$$

f : physical focal length

s_x : each px --- mm wide

1. If the physical focal length remains unchanged but the pixels become smaller, does f_x , measured in pixels, increase or decrease?
2. Two cameras have the same physical focal length and sensor size. Camera A produces a 640×480 image; camera B produces a 1280×960 image using twice as many pixels along each dimension. Which camera should have the larger f_x in pixels?
3. Suppose a point projects 100 pixels to the right of the principal point when $f_x = 500$. If f_x becomes 1000,

with X/Z unchanged, where will it appear relative to the principal point?

4. A subtle one: if f_x increases, does the camera see a wider or narrower portion of the world?

$$1. \uparrow f_x = \frac{f}{s_x \downarrow} \leftrightarrow \text{smaller} \rightarrow f_x \uparrow$$

$$2. f_{xB} = 2f_{xA} \Rightarrow \text{camera B}$$

$$3. u - c_x = \frac{x}{2} \cdot f_x$$

$$100 = 500 \frac{x}{2} \rightarrow 100 \cdot \frac{x}{2} = ? \neq 200 \Rightarrow (200, 0)$$

4. narrower

! larger focal length $\rightarrow$ greater pixel displacement $\rightarrow$ more magnification $\rightarrow$ narrower field of view.

- Increasing physical focal length:
 - Magnifies the image.
 - Moves projected points farther from the principal point.
 - Narrows the physical field of view.
- Making pixels smaller:
 - Increases f_x measured in pixels.
 - Increases image resolution.
 - Does not change the physical field of view when sensor dimensions and lens stay unchanged.
- Physical horizontal field of view depends on sensor width W and physical focal length f :
$$\text{FOV}_x = 2 \tan^{-1} \left(\frac{W}{2f} \right)$$
- Therefore:
 - Longer focal length $\rightarrow$ narrower field of view.
 - Wider sensor $\rightarrow$ wider field of view.
 - Higher pixel count alone $\rightarrow$ same field of view, finer sampling.

> A larger focal length number in pixels does not automatically mean a narrower FOV. Image dimensions in pixels must also be considered.

Camera Intrinsic Matrix

intrinsic: what kind of camera is this

> camera's internal parameters

extrinsics! where is the cam,
where is it looking

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

→ intrinsic matrix! it describes
how the cam. converts normalized
camera coords. into pixels

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} X/Z \\ Y/Z \\ 1 \end{bmatrix}$$

then

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

multiplying gives:

$$u = f_x \cdot \frac{X}{Z} + c_x$$

$$v = f_y \cdot \frac{Y}{Z} + c_y$$

1. Which values inside K control image magnification?
2. Which values translate the coordinate origin from normalized $(0, 0)$ to the principal point?
3. Does K describe the camera's position and rotation in the world, or only its internal projection characteristics?
4. If

$$K = \begin{bmatrix} 800 & 0 & 640 \\ 0 & 800 & 360 \\ 0 & 0 & 1 \end{bmatrix}$$

what are f_x, f_y, c_x, c_y ?

1. $f_x, f_y \Rightarrow f_x$: scales horizontal, f_y :

2. c_x, c_y

3. only its internal projection characteristics $\left(\begin{array}{l} \text{-where camera in the world } x \\ \text{-where dir. } x \\ \text{-how rotated } x \end{array} \right)$

4. 800, 800, 640, 360

→ using K directly with 3D point.

> compact projection equation: $s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$

scale factor = in simple model it becomes Z

$$Y = \begin{bmatrix} 800 & 0 & 640 \\ 0 & 800 & 360 \\ 0 & 0 & 1 \end{bmatrix}, \quad P = \begin{bmatrix} 1 \\ 0.5 \\ 2 \end{bmatrix}$$

$$X = \begin{bmatrix} 2080 \\ 1120 \\ 2 \end{bmatrix} \rightarrow \text{this is a homogeneous pixel coord. not yet the ordinary pixel coord.}$$

$\rightarrow$ we div. by last component $\rightarrow$ because it's Z ; we div. by Z

$$= \begin{bmatrix} 1040 \\ 560 \\ 1 \end{bmatrix} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$



- Increasing physical depth Z while holding X, Y fixed $\rightarrow$ projection moves toward the principal point.
- Scaling the entire homogeneous vector $\rightarrow$ projection does not move.

Lens Distortion.

$\rightarrow$ the ideal pinhole equations assume the lens maps points perfectly
Real lenses bend rays differently depending on where they pass thru the lens

1. Would you expect distortion to be stronger near the principal point or near the image corners? Why?
2. Imagine photographing a perfectly straight grid. Near the image edges, its lines appear curved. Could adjusting only f_x, f_y, c_x, c_y correct that curvature?
3. Why is a chessboard useful for discovering lens distortion?
4. If two pixels are equally far from the principal point but in different directions, would basic radial distortion affect them similarly? What does "radial" suggest here?

1. Distortion generally increases w/ radial distance from the pp.:

$$r^2 = x^2 + y^2$$

Near the prin. point. r is small. Near the corners r is larger

2. additional *non linear distortion parameters that moves pixels by diff amounts depending on their distance from p.p. "needed".

3. we use chessboard's known geometry to measure how much cam. distorts the image.

4. $r^2 = x^2 + y^2$

$$\text{scale} = 1 + k_1 r^2 + k_2 r^4 + k_3 r^6$$

$$\underbrace{x_d = x \cdot \text{scale}}_{\text{distorted}}$$

$$\underbrace{y_d = y \cdot \text{scale}}_{\text{distorted}}$$

} equal distance from the pp
means
similar distortion strength.

- Barrel distortion: outer points are compressed inward; straight lines appear to bulge outward.
- Pincushion distortion: outer points are stretched outward; straight lines bend inward.

Tangential Distortion

> Radial distortion assumes the lens system is perfectly centered and symmetric.

↳ In a real camera it may NOT.

→ p_1, p_2 additional coefficients

it does not depend only on distance from the principal point

OR

does not necessarily move points in/onward

! ideal point → radial distortion + tangential distortion → observed point.

k_1, k_2, k_3 : radial distortion

p_1, p_2 : tangential distortion

$$\Delta x_{TAN} = 2 p_1 xy + p_2 (r^2 + 2x^2)$$

$$\Delta y_{TAN} = p_1 (r^2 + 2y^2) + 2 p_2 xy$$

- Radial distortion models symmetric inward/outward warping around the principal point.
- Tangential distortion models asymmetric warping caused by lens-sensor misalignment.
- Calibration estimates both from observed chessboard corners.
- Undistortion approximately reverses their effects.

1. The whole projection appears horizontally shifted. Which parameter would you first suspect: c_x , k_1 , or p_1 ?
2. Straight lines bend increasingly near every image edge in a roughly symmetric pattern. Which parameter family would you suspect?
3. The warping is noticeably asymmetric between different sides of the image. Which parameter family would you suspect?

1. c_x : shifts the pp horizontally

2. radial distortion

3. tangential distortion

Camera coordinate frame

> OpenCV camera convention:

- O_g : camera's optical center

- $+X$: camera's right

- $+Y$: camera's downward direction

- $+Z$: forward, into the scene (forward from the cam.)

a point: $p_c: \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \rightarrow$ point relative to the cam.

1. An object is directly in front of the camera. What should be approximately true about X_c and Y_c ?

2. The object moves to the camera's right without changing depth. Which coordinate changes, and in which direction?
3. The object moves closer to the camera. Does Z_c increase or decrease?
4. Under the OpenCV convention, an object appears above the principal point. Would you expect Y_c to be positive or negative?

1. both = 0
2. +X changes, increases
3. decreases
4. negative

World Coordinate Frame

> world frame is an external reference that we choose
e.g. chessboard

→ during chessboard calibration
attaching the object frame to the chessboard;

- O_g : one selected corner of chessboard
- X_w : along one row
- Y_w : along one col.
- Z_w : perpendicular to the board

! chessboard is flat $\Rightarrow Z_w = 0$

$$P_c = R P_w + t$$

rotation
between
frames

point
in world
frame

translation
between
origins

1. If we choose a chessboard corner as the world-frame origin, what are that corner's world coordinates?
2. If the chessboard has 25 mm square spacing, what might be the world coordinates of the corner one square to the right of the origin?
3. If we move the entire chessboard, do the printed corners' coordinates relative to the chessboard change?
4. If their chessboard coordinates remain unchanged, what must change in $P_c = R P_w + t$ to describe the board's new position relative to the camera?

board's new position relative to the camera:

1. $(0, 0, 0)$
2. $(25, 0, 0)$
3. NO, the printed corners' coords. not change, cuz coord. frame moves w/ the board
4. R and/or t change
 - ↳ moving $\rightarrow t$ changes
 - rotating $\rightarrow R$ changes

EXAMPLE:

t : where is the chessboard's origin according to the cam.

Suppose the board is parallel to the camera and its origin is:

- 0.50 m to the camera's right
- 0.20 m downward
- 1.00 m in front

Then:

$$t = \begin{bmatrix} 0.50 \\ 0.20 \\ 1.00 \end{bmatrix}$$

Because the board and camera axes are aligned, assume:

$$R = I$$

Take the corner one square right:

$$P_w = \begin{bmatrix} 0.025 \\ 0 \\ 0 \end{bmatrix}$$

Convert it to camera coordinates:

$$P_c = IP_w + t$$
$$P_c = \begin{bmatrix} 0.025 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.50 \\ 0.20 \\ 1.00 \end{bmatrix} = \begin{bmatrix} 0.525 \\ 0.20 \\ 1.00 \end{bmatrix}$$

So the same corner is described differently depending on the reference frame:

- Relative to the board: $(0.025, 0, 0)$
- Relative to the camera: $(0.525, 0.20, 1.00)$

Now move the entire board 0.20 m closer:

$$t_{\text{new}} = \begin{bmatrix} 0.50 \\ 0.20 \\ 0.80 \end{bmatrix}$$

The corner's board coordinate remains:

$$P_w = (0.025, 0, 0)$$

But its camera coordinate becomes:

$$P_c = (0.525, 0.20, 0.80)$$

Rotation Matrices

a 3D rotation matrix: 3×3

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\begin{bmatrix} r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

no rotation: $R = I$; rotating 90° around Z

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The complete idea

$$P_c = RP_w + t$$

Read it as:

1. P_w : start with the corner's address on the chessboard.
2. RP_w : determine which direction that offset points after rotating the board.
3. $RP_w + t$: place the rotated board at its location relative to the camera.
4. P_c : obtain the corner's address according to the camera.

Or even shorter:

- P_w : where is the point on the board?
- R : which way is the board facing?
- t : where is the board's origin?
- P_c : where is the point relative to the camera?

> for every calibration photograph, we have!

P_w , (u,v) , χ , distortion coeff., $R-t$

across multiple photographs:

P_w , χ , distortion coeff. remains the same

$R-t$ changes cuz. the board has a diff. pose each img.

Calibration searches for values that best explain all the detected corners across all photographs.

It then performs a check:

1. Take a known chessboard point P_w .
2. Transform it using R, t .

3. Project it using K and distortion.
4. Predict where the corner should appear.
5. Compare that prediction with the corner actually detected.

That difference is the reprojection error.

EXAMPLE: Predicted corner : (512, 300)
detected corner : (515, 304)

this is the error for one detected corner $\leftarrow e = \sqrt{(512-515)^2 + (300-304)^2} = 5$

> calibration combines errors across

- all chessboard corners
 - all calibration images
- and reports an overall value (RMS reprojection error)

Lower error: predicted corners align more closely w/ detected corners

Higher error: estimated camera model does not explain the observations well

Possible causes of high error include:

- Blurry images
- Incorrect corner detection
- Wrong chessboard dimensions
- Wrong square-size measurement
- Too few images
- Nearly identical board poses
- Poor coverage near the image edges
- Camera focus or zoom changing between images

1. Why should we photograph the chessboard near the image edges and corners, not only at the center?

2. Why should we tilt the chessboard instead of taking every photograph perfectly front-facing?
3. If the camera's zoom or focus changes halfway through capturing images, which supposedly constant camera parameters might change?

1. because lens distortion strongest there
2. Tilting the board reveals how perspective changes w/ orientation and depth
If every img is front-facing, calibration was difficultly separating effects (focal length, distance, board pose)
3. Zoom or focus changes can alter intrinsic parameters

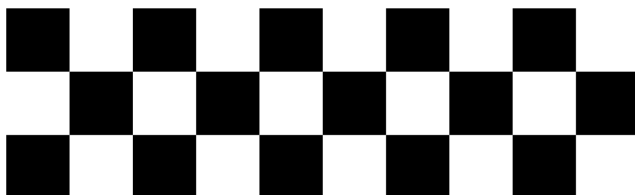
Quantity	Meaning	Across calibration images
P_w	Fixed corner locations on the board	Same
K	Camera intrinsics	Assumed same
Distortion	Lens warping	Assumed same
R, t	Board pose relative to camera	Different
P_c	Corner positions relative to camera	Different
(u, v)	Observed corner pixels	Different

1. A chessboard moves from the image center toward a corner. Which parameters receive better evidence: distortion coefficients or board square size?
2. The chessboard changes angle between two photographs. Which quantities change: K , or R, t ?
3. The lens zoom changes halfway through the dataset. Which "must remain constant" assumption has been broken?

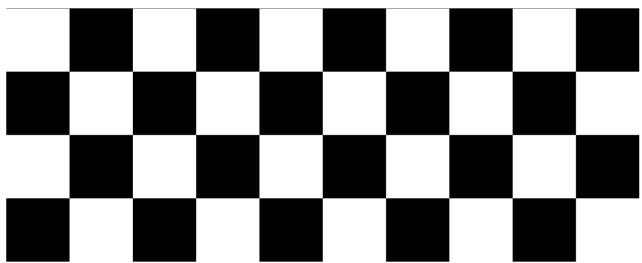
1. distortion coeff. because distortion is strong around corners
2. R, t
3. camera intrinsics

Experiment 1.

1. Object/world points



→ printed board has
 $9 \times 6 = 54$
 inner corners



This is a 4x4
OpenCV chessboard
<https://opencv.org/>

for every corner:

$$P_w = (x_w, y_w, z_w)$$

creates an array w/ shape:

$(54, 3)$ → describes physical chessboard

↳ 54 rows; one row per corner

↳ 3 cols: x_w, y_w, z_w

2. Where the coords. come from

↳ we chose top-left inner corner
as origin: $(0, 0, 0)$

Moving right increases x_w

Moving down increases y_w

$z_w = 0$ becaz chessboard is flat.

→ moving the board changes
the detected
pixel coords.

↳ internal mm
coords. stay
fixed

3. Image points

Open CV detects where those same
corners appear in a photograph
 (u, v)

An image-point array contains 54 detected
pixel locations

4. What changes between photographs

Data	Changes between photos?	Reason
Board coordinates P_w	No	The printed grid never changes

Pixel coordinates (u, v)	Yes	The board appears elsewhere in the image
Camera matrix K	Assumed no	Same camera and resolution
Distortion	Assumed no	Same lens
Board pose R, t	Yes	Board position and orientation change

with 19 photographs:

- 19 copies of the same 54 known board points

- 19 diff. sets of 54 detected pixel points

$\Rightarrow$ Interpreting Calibration

img center:

$$(1920/2, 1080/2) \\ = (960, 540)$$

est. principal point:

$$(c_x, c_y) = (961.55, 534.38)$$

$\rightarrow$ very close to img center so it's reassuring

focal lengths:

$$\left. \begin{aligned} f_x &= 1448.12 \\ f_y &= 1446.35 \end{aligned} \right\} \begin{aligned} &\text{almost identical} \\ &\text{expected for spr. pixels} \end{aligned}$$

$\rightarrow$ the camera matrix:

$$K = \begin{bmatrix} 1448.12 & 0 & 961.55 \\ 0 & 1446.35 & 534.38 \\ 0 & 0 & 1 \end{bmatrix}$$

> the distortion coeff.

$[k_1, k_2, p_1, p_2, k_3]$

Distortion coefficients:

$[-0.023444 \quad 0.067177 \quad -0.001909 \quad -0.000225 \quad -0.129417]$

the RMS error:

0,587 px

! Curved images 03-10 do not show large errors

↳ Low reprojection error does not mean
the physical calibration target was correct.

Because these images helped fit the model, the optimizer can partially absorb target curvature into K , distortion, R , and t . Reprojection error measures how well the fitted model explains its own data—not necessarily physical truth.

⇒ Section 1 - Geometric
Computer Vision

our calibrated camera:

↳ 1920×1080

↳ $f_x \approx 1462,64$

$f_y \approx 1460,17$

↳ principal point:

$(958,35, 531,61)$

↳ RMS reprojection error:

0,456 pxs

=> mild radial and
tangential distortion

1. Why does undistortion change the corners more than the image center?
2. Which are permanent camera properties: K , distortion, R , or t ?
3. Why did the chessboard need to remain physically flat?
4. Why did we capture the board in several positions and orientations?
5. In plain language, what does a reprojection error of approximately 0.456 pixels tell us?

1. distortion is stronger on the corners

2. the internal mm coords stay fixed.
the detected pixel coords change

3. Diverse views make the calibration more robust

4. for reusable results.

↳ K and distortion are reusable

↳ $R - t$ belongs to board pose

5. difference between detected corner pixels and
OpenCV's reprojected predictions